

Sameeksha Mahesh

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Education

University of California, Davis

Sep. 2025 – Mar. 2027

Master of Science, Computer Science

GPA: 4.0/4.0

Relevant Coursework: Machine Learning & Discovery, Design & Analysis of Algorithms, Systems Engineering

Dayanand Sagar College of Engineering

Aug. 2020 – Jun. 2024

Bachelor of Engineering, Computer Science & Engineering

GPA: 3.97/4.0

Technical Skills

Languages	Java, C++/C#, Python, SQL, Go, TypeScript, JavaScript
Frameworks & Libs	Spring Boot, Flask, React, Node.js, Elasticsearch, Streamlit, dbt, Redis, Matplotlib, PyTorch
AI/ML & Agentic	LLMs, LangChain, HuggingFace, Vector Embeddings, RAG Pipelines, ChromaDB, TensorFlow
Data & Streaming	Kafka, Spark (PySpark) Structured Streaming, MLflow, Feast
Developer Tools	Git, Docker, Kubernetes, Jenkins, GitHub Actions, Datadog, Postman, Linux
Databases	MySQL, PostgreSQL, MongoDB, Oracle, Snowflake
Cloud & DevOps	AWS (EC2, Lambda, EBS, S3), Azure, CI/CD Pipelines, REST APIs, Microservices

Experience

University of California, Davis

Jan. 2026 – Mar. 2026

Teaching Assistant — ECS 140A (Programming Languages)

- Led Java OOP and Haskell discussion sections and office hours for 100+ students, delivering targeted feedback on variables, scoping, exceptions, and debugging; designed an **automated code grader** in Python that eliminated manual grading for 100+ students, cutting evaluation turnaround from days to minutes and increasing submission iteration rate.

Perfios Software Solutions Pvt. Ltd

Apr. 2024 – Sep. 2025

Software Development Engineer

- Redesigned RESTful microservices in Java and Spring Boot with caching, response compression, and async request handling, reducing API response times by **20%** across 5+ core payment services.
- Built an intelligent document processing pipeline leveraging Elasticsearch ML-based **vector similarity** to extract and match structured **entity data** from documents, combining **OCR** text extraction with logo recognition to improve automated match rates from 25% to 65% across international bank submissions.
- Shipped a full-stack invoicing platform from scratch (Spring Boot, JavaScript) with **role-based access control (RBAC)**, **AES-256** field-level encryption, and automated financial report generation; deployed to production, replacing a manual reporting workflow for enterprise clients.
- Containerized 8+ microservices with Docker and Kubernetes; automated CI/CD pipelines via GitHub Actions and Jenkins, cutting release cycle time by 25% and deployment failures by 40%.

Brane Enterprises Pvt. Ltd

Jun. 2022 – Jan. 2023

Automation Intern

- Automated **20+ API** test workflows in **Postman** and integrated them into the **CI/CD** pipeline, reducing manual QA effort by **15%** and increasing test coverage by **25%**.
- Optimized 3 client-facing automation modules in **Node.js** across **Agile sprint** cycles, improving average task execution speed by **20%** with **Git**-managed version control.

Projects

Real-Time Fraud Detection Feature Store | [Code](#)

Jul. 2026

- Designed and built a real-time feature-store pipeline: **Kafka** ingestion, **Spark (PySpark) Structured Streaming** (5-minute tumbling windows) computing windowed aggregates, and a dual online (**Redis**) / offline (**MinIO/S3 Parquet**) feature-store split, with the full stack **orchestrated via Docker Compose** across 5+ containerized services.
- Built a **FastAPI** scoring service reading windowed features from Redis at request time to serve sub-second predictions; trained and version-tracked an XGBoost classifier via **MLflow**, and documented production tradeoffs (dual Kafka listeners, S3 partitioning, feature-schema contracts) for reliability at scale.

Agentic RAG Document QA System | [Code](#)

Dec. 2025

- Built an end-to-end **agentic RAG** pipeline from scratch — sentence-aware chunking (5-sentence windows, 2-sentence overlap), Hybrid BM25 + Dense retrieval with Reciprocal Rank Fusion for **grounded context retrieval**, and a **ChromaDB** vector store, achieving 100% retrieval precision on a 23-question held-out benchmark.
- Orchestrated** a two-stage retrieve-and-rerank pipeline (bi-encoder + cross-encoder reranking) with multi-turn conversation memory, LLM-based query rewriting, and streaming generation; improved Token F1 from 0.44 → 0.73 over baseline and deployed as a **FastAPI** REST API with **RAGAs**-style faithfulness scoring and JSONL audit logging for **evaluation and observability**.